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| **PI Name:** | Steinberg, Susan L. Ph.D. | | |
| **Project Title:** | Flow and Distribution of Fluid Phases through Porous Plant Growth Media in Microgravity | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES | | |
| **Program/Discipline--Element/Subdiscipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support | | |
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| **Human Research Program Risks:** | None | | |
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| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Organization Name:** | KBR/NASA Johnson Space Center | | |
| **PI Address 1:** | Human Health and Performance Contract | | |
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| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77058 | **Congressional District:** | 36 |
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| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 4 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | Reddi, Lakshmi ( Kansas State University )<br>Or, Dani ( University of Connecticut )<br>Alexander, J. Iwan D. ( Case Western Reserve University )<br>Daidzic, Nihad ( National Center for Microgravity Research )<br>Jones, Scott ( Utah State University )<br>Tuller, Markus ( University of Idaho )<br>Kluitenberg, Gerard ( Kansas State Unversity )<br>Xiao, Ming ( California State University ) | | |
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| **Task Description:** | NASA's Advanced Life Support program believes that particulate solid substrates are best suited to meet its short and long term needs for plant growth during space flight. Due to mass, volume and power constraints, root modules are characterized by small volumes of growth media and high root density. These conditions leave little margin for error in control of water, air and nutrition. Results from plant growth experiments utilizing particulate growth media during space flight have demonstrated the difficulties associated with providing reliable reproducible air and water (and thus nutrients) supply to plant roots. This is attributed to the general lack of understanding of fundamental transport processes occurring in the growth media. The quantitative characterization of physical processes associated with the flow of wetting and non-wetting phase(s) in particulate plant growth media in microgravity is essential for the successful design and control of plant production systems during space flight. It has recently become evident that the macroscopic dynamic behavior of the temporal and spatial under-saturation levels in particulate media strongly depends on the physical behavior of the fluid at the microscopic (pore-scale) level. Ultimately, the stability and break-up of liquid-bridges, -columns, and -films, dynamic rearrangement of particles, contact angle dynamics and hysteresis, phase discontinuities, capillary fingering and entrapment of the non-wetting phase together with the time-dependent complex geometry of porous media will determine the saturation levels and its stability on macroscopic (integral) scales. However, it is not yet obvious if and how much some of these micro-scale phenomena are gravity dependent.Therefore, we propose a comprehensive program of research that will employ some new advanced experimental, numerical and analytical techniques for examining flow and distribution of fluids in plant growth media. For example, Nuclear Magnetic Resonance Imaging (MRI) will provide a visual record of wetting front dynamics and gas diffusion in porous media. New mathematical models that account for adsorption and liquid films, discontinuous phases, effect of vibration, solitary ganglion, pore network behavior, fingering, capillary instabilities, air entrapment, medium anisotropy and heterogeneity, and the hyperbolic nature of moisture distribution will be evaluated. Upscaling procedures will be employed to describe microscale phenomenon on scales relevant to root modules. Hydraulic conductivity and water retention characteristics will be experimentally verified under simulated microgravity conditions using terrestrial zero-g simulators. The final conceptual model of water and air transport will be verified in 1 g by forward and inverse simulation on the root module scale using flight compatible measurement techniques. The research will result in a model and experimental methods that are ready for final validation in microgravity. Improve control and monitoring of water delivery to plants in microgravity Improve plant growth systems for microgravity crop production and research. Reduce or eliminate environmental control issues. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This work will provide scientific knowledge of fundamental mechanisms and processes related to water and air movement through plant growth media in microgravity. In order to achieve this goal current earth-based conceptual models of flow through porous media will be critically analyzed. We expect this work will make significant contibutions to understanding flow phenomenon on earth. This information will be of interest to hydrologists, agricultural engineers, soil physicists, and agronomists/horticulturists who are interested in water and air transport through soil (porous media) or removal of soil contaminants. |
| **Task Progress:** | Impact of Reduced Gravity Conditions on Fluid Behavior in Plant Growth Media – Challenges<br>In 2002 this NASA funded multidisciplinary team began to address flow and distribution of fluid phases in porous plant growth media in microgravity. This research effort has identified several problems ranging from inadequate characterization of media properties under variable gravity, to phase entrapment and onset of flow behavior not adequately represented by standard flow theories. Some of the key issues involve:<br><br>1. Characterization of porous plant growth media – optimal set points for water delivery in any g environment strongly depend on growth-media hydraulic properties. Although baked ceramic aggregates have been used in space flight for over 10 years, the team conducted the first rigorous characterization of physical and hydraulic properties. This work has enhanced the understanding of aggregate dual porosity, pore size distribution, water conducting porosity, air entrapment, and air entry that allows critical examination of water delivery set points that have been used in space flight. Water retention remains a key physical property of interest with recent KC-135 measurements providing some of the very limited microgravity datasets to date.<br><br>2. Small volumes and susceptibility to heterogeneous fluid distributions - limitations on mass and size of flight-sized root modules accentuate the impact of inherently irregular wetting patterns in particulate media and the role of root volume. Wetting front irregularities may span the entire root module depth used for space flight promoting entrapment and discontinuous phase formation. Root proliferation modifies media properties, especially with high root density where an entirely new porous medium is created with different physical and hydraulic properties. Consequently, water management based on macroscopic criteria (set points) may be constantly changing.<br><br>3. Micro- and reduced gravity effects on water transport - water moves within porous media under the influence of pressure, gravity and surface/capillary forces. Surface/capillary forces dominate in micro- or low gravity environments when both water and air phases are present. Wetting front irregularities or surface forces that cause water and air to segregate into wet and dry clusters are difficult to control or predict. Irregularities in pore size distribution and expansion of capillary length scales enhance phase entrapment in 0g. Once these clusters become established during initial infiltration they may be hard to remove in 0g due to the combination of water-primed pathways and dominance of capillary forces. Additionally infiltration into pre-wetted porous media occurs faster in 0g.<br><br>4. Water content sensing and control - matric potential or water content sensors have been used to control and schedule water delivery to plant growth media. Limited sensing volume of most sensors used in root modules (<1 cm3) combined with discontinuous phases (#2 and 3 above) may result in “accurate” readings on the mesoscale (several pores or particles), but an inaccurate assessment of water content in the bulk volume of growth media.<br><br>Management of Fluids in Plant Growth Media under Reduced Gravity – Opportunities<br><br>The greatest promise for improved monitoring and control of fluids in plant growth media is related to advances in the following areas:<br><br>Engineered plant growth media - Horticulturists spent the better part of the 20th century developing plant growth media for containerized production. Use of baked ceramic aggregates for space flight originated with scientists who had previously used them in ground-based research. Future efforts for microgravity should focus on engineered flight-compatible plant growth media with a prescribed pore size distribution to minimize phase entrapment and |

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| | segregation of liquid and gaseous supply pathways. This will require use of hydrophobic regions, inclusion of slow release nutrients, and an optimized water supply system. For planetary surfaces the presence of gravity and less rigorous size constraints may ease requirements for the precise nature of porous media. The suitability of local regolith alone or in combination with other materials for use in low g environments must be examined. Understanding and controlling multiphase flow through porous media has applications other than plant production, including operation of bioreactors proposed for water revitalization. Sensing and control - better understanding of sensor operation and the use of engineered media with predictable and controllable hydraulic properties may improve sensing of media water status. However, the presence of roots may change these properties over time. New sensing strategies need to be developed that are compatible with properties of dense root zones.<br><br>Root zones and rhizosphere - better understanding of the effects of roots on water and air transport in small volumes of growth media is essential. Roots, root exudates and rhizosphere microfauna activity may change the physical and hydraulic properties of growth media over time and must also be considered in engineered media. |
| **Bibliography Type:** | Description: (Last Updated: 03/14/2017) |
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